CS 2840 Fall 2025 Student Name: Yuyuan Chen, Haichuan Wang

Gromov-Wasserstein Alignment: Statistics, Computation, and Geometry

1 Recall: Optimal Transport

We briefly review the Kantorovich formulation of Optimal Transport (OT) and the induced Wasserstein geometry.

1.1 Kantorovich Optimal Transport

Let $(\mathcal{X}, d)$ be a metric space and let μ, ν be probability measures on $\mathcal{X}$. The *Kantorovich optimal transport problem* is

$$W_p(\mu,\nu)^p \;=\; \inf_{\pi\in\Pi(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{X}} d(x,y)^p \, d\pi(x,y), \tag{1}$$

where $\Pi(\mu,\nu)$ denotes the set of couplings of μ and ν:

$$\Pi(\mu,\nu) = \Big\{\pi \geq 0 :\; \pi(\cdot,\mathcal{X}) = \mu,\; \pi(\mathcal{X},\cdot) = \nu\Big\}.$$

This is a linear program in the variable π, with linear marginal constraints and linear objective.

1.2 The Wasserstein Distance

The quantity W_p defined above is the *p-Wasserstein distance*. A key property is that it *lifts* the base metric d: for any $x, y \in \mathcal{X}$,

$$W_p(\delta_x, \delta_y) = d(x,y), \tag{2}$$

so the Wasserstein space of probability measures contains $\mathcal{X}$ as a subset via Dirac masses.

1.3 Geodesics in Wasserstein Space

The space $(\mathcal{P}_p(\mathcal{X}), W_p)$ is a geodesic metric space. Given an optimal coupling $\pi^\star$ between μ_0 and μ_1, the *displacement interpolation* $(\mu_t)_{t\in[0,1]}$ is defined by pushing $\pi^\star$ forward along the constant-speed geodesics in $\mathcal{X}$:

$$\mu_t = ((1-t)x + ty)_{\#}\pi^\star.$$

This satisfies

$$W_p(\mu_s,\mu_t) = |t-s| \cdot W_p(\mu_0,\mu_1),$$

so (μ_t) is a constant-speed geodesic in Wasserstein space.

1.4 Barycenters

Given measures $\mu_1,\ldots,\mu_N$ and nonnegative weights $w_1,\ldots,w_N$ with $\sum_i w_i = 1$, the *Wasserstein barycenter* is defined as

$$\mu^\star \;\in\; \arg\min_{\rho\in\mathcal{P}_p(\mathcal{X})} \sum_{i=1}^{N} w_i W_p(\rho,\mu_i)^p. \tag{3}$$

Barycenters generalize Euclidean means to the metric space $(\mathcal{P}_p(\mathcal{X}), W_p)$ and play a central role in applications such as shape analysis, clustering, and generative modeling.

1.5 Application: Mixed Style Transfer

A classical OT application views images as probability distributions. To combine multiple style images $\mu_{S_1}, \ldots, \mu_{S_N}$ with weights w_i, one computes their Wasserstein barycenter

$$\mu^*_{\text{style}} = \arg\min_{\rho} \sum_{i=1}^{N} w_i W_2(\rho, \mu_{S_i})^2.$$

Then one interpolates between a content image distribution μ_C and the style barycenter along the Wasserstein geodesic:

$$\mu_t = \text{Geo}_t(\mu_C, \mu^*_{\text{style}}).$$

This viewpoint interprets style transfer as barycenter computation followed by geodesic interpolation. Details can be seen in [1].

1.6 Problems with the OT framework

There are several problems with the OT framework.

- Structure-preserving interpolation: naive Wasserstein transfer doesn't incentive to look at and preserve local structure
- Naive OT/Wasserstein distance cannot quantify discrepancy between incompatible spaces

These problems motivate to use Gromov-Wasserstein distance.

2 Gromov-Wasserstein Alignment

In many applications, we are given two heterogeneous or structured datasets and we wish to quantify how similar they are, or to compute an alignment between them, however, sometimes there is not a notion of distance between points in these two spaces.

A natural formalism for this problem is to treat each dataset as a *metric measure space*

$$(X, d_X, \mu) \qquad \text{and} \qquad (Y, d_Y, \nu),$$

where d_X and d_Y are intrinsic distances and μ, ν are probability measures over the two domains. The goal is to find a matching between the two datasets that preserves geometry as much as possible. Formally, for a transport map $T : X \to Y$, we would like pairwise distances to be preserved:

$$d_X(x_i, x_j)^2 \approx d_Y\big(T(x_i), T(x_j)\big)^2,$$

so that the *distance distortion* $|\, d_X(x_i, x_j)^2 - d_Y(T(x_i), T(x_j))^2|$ (i.e. informally, how much T fails to be an isometry) is small.

2.1 Gromov–Wasserstein Distance

Since no canonical cross-space distance $d(x, y)$ exists, classical OT cannot be used directly. Instead, one uses the *Gromov–Wasserstein* (GW) distance, which compares *intra-space structures* by aligning pairwise distances.

The (squared) Gromov–Wasserstein distance is defined as

$$GW(\mu, \nu) = \inf_{\pi} \left(\mathbb{E}_{(X,Y),(X',Y') \sim \pi \otimes \pi} \left[\left| \, d_X(X, X')^2 - d_Y(Y, Y')^2 \, \right|^2 \right] \right)^{1/2}. \tag{4}$$

The coupling π is a joint distribution between X and Y, and $\pi \otimes \pi$ corresponds to two independent draws from π. This quadratic dependence on π reflects the fact that GW compares *pairs* of points.

More generally, for exponents $p, q \geq 1$ one defines

$$GW_{p,q}(\mu, \nu) = \inf_{\pi} \left(\mathbb{E}_{\pi \otimes \pi} \left[\, | \, d_X(X, X')^q - d_Y(Y, Y')^q \, |^p \right] \right)^{1/p}. \tag{5}$$

Quadratic difficulty. In contrast to classical OT, where the cost is linear in the coupling, GW has a *quadratic* structure. This significantly complicates both analysis and computation.

Properties of the Gromov-Wasserstein distance

- Finiteness. If μ and ν satisfy
$$\mathbb{E}[d_X(X,X')^{pq}] < \infty, \qquad \mathbb{E}[d_Y(Y,Y')^{pq}] < \infty,$$
then $GW_{p,q}(\mu,\nu) < \infty$.
- Identification (Isometry Invariance). One has
$$GW_{p,q}(\mu,\nu) = 0 \quad \Longleftrightarrow \quad \exists \text{ an isometry } T : X \to Y \text{ such that } T_{\#}\mu = \nu.$$
Thus $GW_{p,q}$ metrizes the space of metric-measure spaces up to isometry.

2.2 Duality for Quadratic GW Distance

We consider the $(2,2)$–Gromov–Wasserstein distance between metric–measure spaces
$$(\mathbb{R}^{d_x}, \|\cdot\|, \mu) \qquad \text{and} \qquad (\mathbb{R}^{d_y}, \|\cdot\|, \nu),$$
and assume that both measures have finite fourth moments:
$$M_4(\mu) := \int \|x\|^4 \, d\mu(x) < \infty, \qquad M_4(\nu) < \infty.$$

The quadratic GW distance is defined as
$$\mathrm{GW}(\mu,\nu)^2 = \inf_{\pi \in \Pi(\mu,\nu)} \iint_{\mathbb{R}^{d_x}\times\mathbb{R}^{d_y}} \iint_{\mathbb{R}^{d_x}\times\mathbb{R}^{d_y}} \left(\|x-x'\|^2 - \|y-y'\|^2\right)^2 d\pi(x,y)\, d\pi(x',y').$$

Decomposition Assume without loss of generality that both measures are centered:
$$\int x \, d\mu(x) = 0, \qquad \int y \, d\nu(y) = 0,$$
(using translational invariance of GW). Then the GW distance admits the following decomposition:
$$\mathrm{GW}(\mu,\nu)^2 = S_1(\mu,\nu) + S_2(\mu,\nu).$$
with
$$S_1(\mu,\nu) = \iint \|x-x'\|^4 \, d\mu(x)\, d\mu(x') + \iint \|y-y'\|^4 \, d\nu(y)\, d\nu(y') - 4 \int \|x\|^2 \, d\mu(x) \int \|y\|^2 \, d\nu(y).$$
$$S_2(\mu,\nu) = \inf_{\pi \in \Pi(\mu,\nu)} \left[\int -4\,\|x\|^2\|y\|^2 \, d\pi(x,y) \; - \; 8 \sum_{i=1}^{d_x} \sum_{j=1}^{d_y} \left(\int x_i y_j \, d\pi(x,y) \right)^2 \right].$$

3 Estimation from Data

Suppose the underlying probability measures μ and ν on metric spaces are unknown. Instead, we observe i.i.d. samples
$$X_1, \ldots, X_n \sim \mu, \qquad Y_1, \ldots, Y_n \sim \nu.$$

- Empirical measures:
$$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}, \qquad \hat{\nu}_n := \frac{1}{n} \sum_{i=1}^{n} \delta_{Y_i}.$$
- Approximate via plug-in
$$\mathrm{GW}(\mu,\nu) \quad \approx \quad \mathrm{GW}(\hat{\mu}_n, \hat{\nu}_n)$$

Asymptotic Consistency If μ and ν have finite fourth moments, then

$$\mathrm{GW}(\hat{\mu}_n, \hat{\nu}_n) \;\longrightarrow\; \mathrm{GW}(\mu, \nu) \qquad \text{almost surely,}$$

as $n \to \infty$. This was established in work by Mémoli (2011). The rate at which $\mathbb{E}\big[\,\big|\,\mathrm{GW}(\mu,\nu) - \mathrm{GW}(\hat{\mu}_n, \hat{\nu}_n)\,\big|\,\big]$ decays is still not known.

We now describe the first known non-asymptotic upper bounds for estimating the Gromov–Wasserstein distance from samples.

Theorem 1 (Zhang–G.-Mroueh–Sriperumbudur, 2024; Ann. of Stat.). *Let $(\mu,\nu) \in \mathcal{P}(\mathbb{R}^{d_x}) \times \mathcal{P}(\mathbb{R}^{d_y})$ be probability measures with compact support of diameter at most $R > 0$. Let $\hat{\mu}_n$ and $\hat{\nu}_n$ be the corresponding empirical measures based on n i.i.d. samples from each distribution. Then*

$$\mathbb{E}\big[\,\big|\,GW(\mu,\nu)^2 - GW(\hat{\mu}_n, \hat{\nu}_n)^2\,\big|\,\big] \;\;\leq\;\; c_{d_x,d_y,R}\left(R^4 n^{-1/2} \;+\; (1+R^4)\, n^{-\frac{2}{(d_x \wedge d_y) \vee 4}} (\log n)^{\mathbf{1}_{\{d_x \wedge d_y = 4\}}}\right).$$

The gist of the proof is that the dual formulation shows that, despite the quadratic structure of GW, its sample complexity behaves like that of a family of linear OT problems indexed by a finite–dimensional parameter A.

4 Limit distribution theory (teaser)

The nonasymptotic rates above describe how fast the empirical GW distance converges in expectation. A natural next question is whether one can obtain *limit distributions* for suitably rescaled empirical GW distances, in the spirit of classical central limit theorems.

The recent work of Rioux–Goldfeld–Kato develops such limit laws in several settings (discrete, semi-discrete, and entropically regularized GW), answering this question affirmatively under appropriate regularity conditions. Roughly speaking, one considers scales r_n such that

$$r_n\big(\mathrm{GW}(\hat{\mu}_n, \hat{\nu}_n) - \mathrm{GW}(\mu,\nu)\big) \;\Rightarrow\; Z,$$

where Z is a nondegenerate limiting random variable.

The proof strategy follows a now standard recipe in modern empirical process theory:

1. **Empirical process convergence.** Using the dual/variational formulation of GW, one can characterize $\mathrm{GW}(\mu,\nu)$ as the value of an optimization problem over a class of dual potentials (or cost perturbations). The empirical counterpart replaces expectations under (μ,ν) with expectations under $(\hat{\mu}_n, \hat{\nu}_n)$. One shows that the associated empirical process converges weakly in the space $\ell^\infty(\mathcal{F})$ for an appropriate function class $\mathcal{F}$.

2. **Gâteaux differentiability.** One then studies the functional

   $$(\mu,\nu) \longmapsto \mathrm{GW}(\mu,\nu)$$

   as a map on the space of probability measures. Under suitable regularity assumptions, GW is Gâteaux differentiable with respect to weak perturbations of (μ,ν), and its derivative is Lipschitz with respect to the sup–norm on the empirical process index set.

3. **Extended functional delta method.** Combining the previous two ingredients with a functional delta method yields the desired limit:

   $$r_n\big(\mathrm{GW}(\hat{\mu}_n, \hat{\nu}_n) - \mathrm{GW}(\mu,\nu)\big) \;\Rightarrow\; D'_{\mu,\nu}\big(G_\mu, G_\nu\big),$$

   where (G_μ, G_ν) is the Gaussian limit of the empirical process, and $D'_{\mu,\nu}$ denotes the Gâteaux derivative of the GW functional at (μ,ν).

For more details, see, e.g., [3].

5 Computation via entropic Gromov–Wasserstein

We now turn to the computational side of Gromov–Wasserstein alignment.

5.1 GW as a quadratic assignment problem

In the discrete setting, suppose we have two empirical measures supported on n points each:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad \nu = \frac{1}{n}\sum_{i=1}^{n} \delta_{y_i}.$$

The 2–2 GW distance between $(\{x_i\}, d_X)$ and $(\{y_i\}, d_Y)$ can be written as

$$\mathrm{GW}(\mu,\nu)^2 = \frac{1}{n^2}\min_{\sigma\in S_n}\sum_{i,j=1}^{n}\Big(d_X(x_i,x_j)^2 - d_Y(y_{\sigma(i)},y_{\sigma(j)})^2\Big)^2,$$

where S_n is the set of permutations. This is a *quadratic assignment problem* (QAP), a well-known NP-complete combinatorial optimization problem. Thus computing GW exactly is intractable in general, and one typically resorts to heuristic or approximate algorithms. Early approaches are based on projected gradient or mirror descent schemes that alternate between updating the coupling and re-evaluating the quadratic cost.

5.2 Entropic Gromov–Wasserstein

A powerful way to obtain both numerical stability and faster algorithms is to introduce entropic regularization, in analogy with entropic OT. The entropic GW distance is defined as

$$\mathrm{GW}_\varepsilon(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)}\left\{\mathbb{E}_{\pi\otimes\pi}\left[(\|X-X'\|^2 - \|Y-Y'\|^2)^2\right] + \varepsilon\,\mathrm{KL}(\pi\,\|\,\mu\otimes\nu)\right\},$$

where $\varepsilon > 0$ is the regularization parameter, and KL denotes Kullback–Leibler divergence. When ε is large, the regularization dominates and the problem becomes smoother and more convex; when ε is small, one recovers an approximation of the true (unregularized) GW distance.

The regularized problem can be rewritten using the $S_1 + S_2$ decomposition as

$$\mathrm{GW}_\varepsilon(\mu,\nu) = S_1(\mu,\nu) + \min_{A\in\mathcal{D}_{M_{\mu,\nu}}} \mathrm{EOT}_\varepsilon^A(\mu,\nu) =: \Phi(A),$$

where $\mathrm{EOT}_\varepsilon^A$ is an entropically regularized OT cost with a linear cost c_A and entropy parameter ε. The optimization now takes place over the finite–dimensional matrix A in the convex set $\mathcal{D}_{M_{\mu,\nu}}$.

5.3 Smoothness and convexity of the objective

A central result of Rioux–Goldfeld–Kato is a detailed smoothness and convexity analysis of the objective $A \mapsto \Phi(A)$:

- Φ is L–smooth on $\mathcal{D}_{M_{\mu,\nu}}$, meaning that its gradient is Lipschitz with constant

  $$L \le 64 \vee \big(32^2\varepsilon^{-1}M_4(\mu)M_4(\nu) - 64\big),$$

  where $M_4(\cdot)$ denotes the fourth moment.

- For sufficiently large regularization, $\varepsilon > 16M_4(\mu)M_4(\nu)$, the function Φ is strictly convex on $\mathcal{D}_{M_{\mu,\nu}}$. This is shown by bounding the eigenvalues of the Hessian:

  $$64 - 32^2\varepsilon^{-1}M_4(\mu)M_4(\nu) \le \lambda_{\min}\big(D^2\Phi(A)\big) \le \lambda_{\max}\big(D^2\Phi(A)\big) \le 64.$$

These properties open the door to applying accelerated gradient methods with global convergence guarantees to the entropic GW problem.

5.4 Fast gradient method with an inexact oracle

The gradient of Φ at A can be expressed in terms of the optimal entropic OT coupling π_A associated with A:

$$\nabla\Phi(A) = 64A - 32\sum_{i,j=1}^{n} x_i y_j^\top \pi_A(i,j),$$

where the sum is taken over sample points for a discrete approximation.

In practice, π_A is computed only approximately via a finite number of Sinkhorn iterations, yielding an *inexact* coupling $\hat{\pi}_A$ that satisfies

$$\|\pi_A - \hat{\pi}_A\|_\infty \le \delta.$$

Replacing π_A by $\hat{\pi}_A$ produces an approximate gradient $\widehat{\nabla\Phi}(A)$, which is then used in an accelerated projected gradient scheme on $\mathcal{D}_{M_{\mu,\nu}}$.

Under appropriate control of the inexactness parameter δ across iterations, one can show that the resulting iterates B_k satisfy the global convergence guarantee

$$\Phi(B_k) - \Phi(B^\star) \le \frac{2L\|B^\star\|_F^2}{(k+1)(k+2)} + O(M),$$

where $B^\star$ is a minimizer of Φ and M collects the accumulated approximation error from the inexact gradients. In the convex regime this matches the optimal $O(1/k^2)$ rate for smooth constrained optimization, while in the general nonconvex case one still obtains convergence to a stationary point.

Overall, this yields a numerically efficient algorithm whose per-iteration cost is dominated by $O(N^2)$ Sinkhorn steps (for N support points) and which achieves significantly better wall-clock performance than mirror descent methods that require repeated $O(N^3)$ cost updates.

6 Numerical results

The numerical experiments compare several algorithms for computing entropic GW distances as the number of points N and the dimension d vary.

- The fast gradient method of Rioux–Goldfeld–Kato exhibits nearly linear convergence in the number of iterations, and its total runtime scales roughly like

  $$\text{time} \approx k \times O(N^2),$$

  where k is the iteration count. This comes from the fact that the cost matrix need not be recomputed at each step; only Sinkhorn iterations are required.

- In contrast, mirror descent approaches that update a full quadratic cost tensor at each iteration incur a much higher per-iteration cost, leading to runtimes of order $k \times O(N^3)$ or $k \times d \times O(N^2)$ depending on the implementation.

Empirically, the fast gradient method achieves the same target value of the entropic GW objective in substantially less time, especially in higher dimensions and for large sample sizes.

7 Gromov–Wasserstein geometry and gradient flows

We now move beyond finite-sample computation and study the *geometry* induced by Gromov–Wasserstein distances.

7.1 Inner-product GW distance

When the two metric measure spaces coincide, i.e. we consider two probability measures μ and ν on the same Euclidean space $\mathbb{R}^d$ with the usual squared-distance cost, one can define an "inner-product" GW distance (IGW) by replacing squared distances with inner products:

$$\mathrm{IGW}(\mu,\nu)^2 = \inf_{\pi\in\Pi(\mu,\nu)} \mathbb{E}_{\pi\otimes\pi}\big[\langle X,X'\rangle - \langle Y,Y'\rangle\big]^2.$$

Here the goal is to align the covariance structure (and more generally the Gram matrix) of the two distributions rather than their pairwise distances. This is particularly natural when one is interested in global structure such as principal directions or shapes.

7.2 Proximal point (JKO) scheme for IGW gradient flows

Given a functional $\mathcal{F}\colon \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ that is invariant under orthogonal transformations, Zhang–Goldfeld–Greenewald–Mroueh–Sriperumbudur introduce an IGW-based Jordan–Kinderlehrer–Otto (JKO) scheme. Starting from an initial measure μ_0, they define a sequence $(\mu_t)_{t\geq 0}$ by

$$\tilde{\mu}_{t+1} = \arg\min_{\mu}\Big\{\mathcal{F}(\mu) + \frac{1}{2\tau}\,\mathrm{IGW}(\mu,\mu_t)^2\Big\},$$

followed by an orthogonal "covariance alignment" step

$$\mu_{t+1} = O_{\mu_t,\tilde{\mu}_{t+1}}\#\tilde{\mu}_{t+1},$$

where $O_{\mu_t,\tilde{\mu}_{t+1}}$ is an orthogonal map chosen to best align the covariances of μ_t and $\tilde{\mu}_{t+1}$. Intuitively, each step decreases the functional $\mathcal{F}$ while making the smallest possible change to the global structure of the distribution, as measured by IGW.

As the time step $\tau \to 0$, the piecewise-constant interpolant of this scheme converges to a continuous curve $(\mu_t)_{t\geq 0}$ in Wasserstein space that solves a nonlinear continuity equation of the form

$$\partial_t \mu_t = -\mathrm{div}(\mu_t v_t), \qquad v_t = -\mathcal{L}^{-1}_{\Sigma_t,\mu_t}\nabla\delta\mathcal{F}(\mu_t).$$

Here $\Sigma_t = \int xx^\top d\mu_t(x)$ is the covariance matrix, $\delta\mathcal{F}(\mu_t)$ denotes the first variation, and $\mathcal{L}_{\Sigma_t,\mu_t}$ is a "mobility" operator which links the IGW gradient to the classical W_2 gradient via

$$\nabla_{W_2}\mathcal{F}(\mu) = \nabla\delta\mathcal{F}(\mu), \qquad \nabla_{\mathrm{IGW}}\mathcal{F}(\mu) = \mathcal{L}^{-1}_{\Sigma,\mu}\nabla_{W_2}\mathcal{F}(\mu).$$

This induces a Riemannian metric on the space of probability measures,

$$g_\mu(v,w) := \langle v,\ \mathcal{L}_{\Sigma_\mu,\mu}w\rangle_{L^2(\mu,\mathbb{R}^d)},$$

which can be viewed as a GW-type deformation of the usual W_2 geometry.

8 Illustrative examples

The slides contain several numerical experiments that illustrate the qualitative differences between Wasserstein and IGW interpolations and flows.

Ellipse interpolation. When interpolating between two point clouds sampled from ellipses with different orientations, the Wasserstein geodesic tends to deform the shape by transporting mass along straight paths in $\mathbb{R}^2$. In contrast, IGW interpolation preserves the elliptical structure more faithfully, performing an almost rigid rotation of the cloud rather than distorting it.

Cat shape interpolation. A similar phenomenon appears in the "cat" example: IGW yields interpolations that preserve the global silhouette of the cat while smoothly adjusting local features, whereas Wasserstein interpolation can introduce undesirable blurring or local distortions. When combined with GW, one can even transport a cat shape into a different target domain while preserving large-scale structure more effectively than pure Wasserstein transport.

IGW gradient flow of entropy. The final experiment compares the gradient flow of the entropy functional under Wasserstein and IGW geometries. Under W_2, the flow resembles standard heat diffusion that quickly smooths out the distribution. Under IGW, the flow decomposes into a "descent" component (which decreases entropy) and a "damping" component (which maintains global structure), leading to markedly different trajectories in distribution space.

These examples highlight that GW/IGW-based geometries provide a complementary view to classical Wasserstein geometry, particularly in tasks where preserving or aligning global structures is more important than pointwise transport.

9 Summary and references

Gromov–Wasserstein distances extend optimal transport to settings where the two datasets live in different metric spaces or even lack a natural ambient space altogether. They quantify the discrepancy between metric-measure spaces, enable alignment of heterogeneous structured data, and induce a rich geometric structure with associated gradient flows. At the same time, their quadratic and nonconvex nature poses significant challenges in statistics and computation.

The works discussed in this lecture develop:

- dual and variational formulations that connect GW to (entropic) OT;
- sharp nonasymptotic rates and limit distributions for empirical GW;
- efficient algorithms with convergence guarantees for entropic GW;
- a GW-based Riemannian structure and gradient flows on spaces of probability measures.

Acknowledgement: Due to the formidable mathematical density of this guest lecture, GPT was used to assist with preparing this scribe, in particular with helping with unpacking and understanding the materials in sections 4, 5, and 7.

References

[1] Youssef Mroueh. Wasserstein style transfer. *arXiv preprint arXiv:1905.12828*, 2019.

[2] Gabriel Rioux, Ziv Goldfeld, and Kengo Kato. Entropic gromov-wasserstein distances: Stability and algorithms. *Journal of Machine Learning Research*, 25:1–52, 2024. arXiv:2306.00182.

[3] Gabriel Rioux, Ziv Goldfeld, and Kengo Kato. Limit laws for gromov–wasserstein alignment with applications to testing graph isomorphisms. *arXiv preprint arXiv:2410.18006*, 2024.

[4] Zhengxin Zhang, Ziv Goldfeld, Youssef Mroueh, and Bharath K. Sriperumbudur. Gromov–wasserstein distances: Entropic regularization, duality and sample complexity. *Annals of Statistics*, 52(4):1616–1645, 2024. doi: 10.1214/24-AOS2406. arXiv:2212.12848.

[5] Zhengxin Zhang, Ziv Goldfeld, Kristjan Greenewald, Youssef Mroueh, and Bharath K. Sriperumbudur. Gradient flows and riemannian structure in the gromov–wasserstein geometry. *Foundations of Computational Mathematics*, 2025. doi: 10.1007/s10208-025-09722-w. arXiv:2407.11800.